Dr. Prathmesh Bhadane

Indian Institute of Technology Bombay, India | prathmesh1378@gmail.com | +91 9503301033 | Google Scholar

Institute Postdoctoral Fellow at IIT Bombay, developing metal-organic framework (MOF) based materials for critical raw material and rare earth element recovery, heavy metal remediation, and heterogeneous catalysis for gas conversion. Research spans MOF synthesis and functionalization, circular repurposing of spent adsorbents as catalysts, and translating lab-scale recovery chemistry into scalable, sustainable-by-design materials.

EDUCATION

Indian Institute of Technology, Gandhinagar **2021 – 2026**

PhD in Materials Engineering; CPI: 8.84/10

- Commendation for Outstanding Innovation (PhD), awarded at the 15th Convocation.

Indian Institute of Technology, Gandhinagar **2019 – 2021**

M.Tech. in Materials Science and Engineering; CPI: 8.00/10.

Mumbai University **2014 – 2018**

B.E. in Mechanical Engineering; 9.18/10

- Secured first position in the second year of engineering with a 9.83/10 CGPA.

EXPERIENCE

MOF-Based Heterogeneous Catalysts for Gas Conversion **Feb. 2026 – Present**

Institute Postdoctoral Fellow, IIT Bombay; Supervisor: Prof. Deepti Kalsi

- Designing MOF-based heterogeneous catalysts for the conversion of CO_2 and CH_4 into value-added chemicals, including formate, methanol, and acetic acid.
- Teaching assistant for CH-831L Instrumentation Laboratory, Department of Chemistry, IIT Bombay.

MOF-Based Environmental Remediation and Resource Recovery **Sept. 2024 – Jan. 2025**

Visiting PhD Scholar, University of Birmingham; Supervisors: Prof. Iseult Lynch and Dr. Swaroop Chakraborty

Researched Safe and Sustainable by Design MOF beads for selective entrapment of rare earth elements, developing MOF-polymer composite beads for sustainable REE recovery.

- Funded by the IITGN Overseas Research Experience Fellowship (6,800 USD).

Sustainable Nanocomposites and MOF Systems for Remediation and Resource Recovery **2021 – 2026**

Doctoral Research, IIT Gandhinagar; Thesis Supervisor: Prof. Abhijit Mishra

- Developed 2D and nanosheet Cu-imidazolate MOFs for selective heavy-metal capture and rare earth element recovery from electronic waste and aqueous streams.
- Built cellulose-acetate–nanoMOF composite beads for scalable lead remediation in complex water systems.
- Filed an Indian patent for an adsorbent and process for rare earth element recovery.

TEACHING AND MENTORING

- Independent Instructor, Materials Characterization Techniques, IIT Gandhinagar (Jan-May 2024).
- Teaching Assistant, Engineering Graphics (undergraduate), IIT Gandhinagar.
- Research supervision: guided Mr. Adit Rambhia (B.Tech, IIT Gandhinagar) on “Development of a Reusable Semi-IPN Polymer-Coated Cellulose Acetate Bead Filtration System for Real-World Antibacterial Applications” (2024-2025).
- Assistant Teacher, Shifa’s Group Tuitions, Ambarnath, Thane (2014–2018) - Mathematics, Classes XI–XII.
- Private Tutor, Mechanical Engineering (2016–2018) - Strength of Materials, Heat Transfer, Fluid Mechanics, Machine Design, for a batch of 40 undergraduates.

PEER-REVIEWED PUBLICATIONS

17 peer-reviewed publications spanning MOF-based environmental remediation, critical mineral recovery, and sustainable materials design, MOF designing and its transformation, heterogeneous catalysis

1. **Bhadane, P.,** Menon, D., Goyal, P., Alizadeh Kiapi, M.R., Satpathy, B.K., Lanza, A., Mikulska, I., Scatena, R., Michalik, S., Mahato, P., Asgari, M., Chen, X., Chakraborty, S., Mishra, A., Lynch, I., Fairén-Jiménez, D., Misra, S.K. A Two-Dimensional Metal-Organic Framework for Efficient Recovery of Heavy and Light Rare Earth Elements from Electronic Wastes. Separation and Purification Technology (2024), 130946. IF: 9.1.

2. **Bhadane, P.**, Dhumal, P., Brun, E., Britton, A., Lynch, I., Chakraborty, S. Safe and Sustainable by Design MOF Beads for Selective Entrapment and Recovery of Rare Earth Elements. Environmental Science & Technology (2025). IF: 12.2.
3. **Bhadane, P.**, Chakraborty, S. Cross-Material Synergies of Carbon Nanomaterials, MOFs, and COFs: Innovative Approaches for Sustainable Environmental Remediation and Resource Recovery. Coordination Chemistry Reviews (2025). IF: 25.6.
4. Dhumal, P.*, **Bhadane, P.***, Ibrahim, B., Chakraborty, S. **(*equal contribution)** Evaluating the Path to Sustainability: SWOT Analysis of Safe and Sustainable by Design Approaches for Metal-Organic Frameworks. Green Chemistry (2025). IF: 10.6.
5. Menon, D., **Bhadane, P**., Goyal, P., Mahato, P., Chakraborty, S., Lynch, I., Mishra, S. An End-to-End Workflow for Selective Heavy Metal Capture and Rare-Earth Element Recovery Using Metal–Organic Frameworks. **Nature Protocols** (2026, just accepted). IF: 16.0.
6. **Bhadane, P.**, Kalsi, D. Rethinking MOFs for REE Recovery: Role of Functional Groups, Selectivity Limitations and Possible Future Directions. Coordination Chemistry Reviews (2026). IF: 25.6.
7. **Bhadane, P.**, Prajapati, D.G., Rambhia, A., Dotiyal, M., Pandey, P.K., Rajput, D., Mishra, A. Antibacterial Activity of Cellulose Acetate–Graphene Oxide Composites: A Comprehensive Multimethod Assessment. ACS Omega (2025). IF: 5.2.
8. **Bhadane, P.**, Mahato, P., Menon, D., Satpathy, B.K., Wu, L., Chakraborty, S., Goyal, P., Lynch, I., Misra, S.K. Hydrolytically Stable Nanosheets of Cu–Imidazolate MOF for Selective Trapping and Simultaneous Removal of Multiple Heavy Metal Ions. Environmental Science: Nano (2024). IF: 5.6.
9. **Bhadane, P.**, Chakraborty, S. Cellulose Acetate-nanoMOF Beads: A Safe, Sustainable and Scalable Solution for Lead Remediation in Complex Water Systems. Environmental Science: Nano (2025). IF: 5.6.
10. **Bhadane, P.**, Mishra, A. The Effect of Alkali Treatment on Pineapple Leaf Fibers (PALF) on the Performance of PALF Reinforced Rice Starch Biocomposites. Journal of Natural Fibers (2022): 1–15. IF: 3.1.
11. **Bhadane, P.**, Mishra, A. Versatile, Flexible Rice Starch–Graphene Oxide Bionanocomposites. Environmental Science: Water Research & Technology (2024). IF: 3.5.
12. Chakraborty, S., **Bhadane, P.**, Wilgner, S.,, Hiroto, K., Ana, B., Sang, P., Superb, M., and Iuliia, M. "Freeze-drying enables resource-efficient isolation of copper imidazolate metal–organic framework nanosheets for transformation-aware lead capture." Green Chemistry (2026). IF: 10.6.
13. Chakraborty, S., Drexel, R., **Bhadane, P.,** Langford, N., Dhumal, P., Meier, F., Lynch, I**.** Integrated Multimethod Approach for Size-Specific Assessment of Potentially Toxic Element Adsorption onto Micro- and Nanoplastics: Implications for Environmental Risk. Nanoscale (2025). IF: 5.1.
14. Chakraborty, S., Mikulska, I., Boseley, R., Pham, S., **Bhadane, P.**, Dhumal, P., Majumder, S. et al. Hierarchical Environmental Exposure Transforms Zeolitic Imidazolate Framework-8 and Increases Toxicity in Daphnia magna. ACS Nano (2026). IF: 17.3.
15. Chakraborty, S., Guilherme Buzanich, A., **Bhadane, P.,** Kitaguchi, H., Pham, S., Mikulska, I**.** Structural Persistence Masks Commit-Point Chemical Transformation in Copper–Imidazolate Nanosheet Metal–Organic Frameworks. ACS Nano (2026). IF: 17.3.
16. Saran, R., Saxena, S., Acharya, H., **Bhadane, P.**, Taki, K. Utilization of Water Treatment Plant Sludge for Creating Green Bricks and Examining Its Gamma Radiation Shielding Potential. Journal of Hazardous, Toxic, and Radioactive Waste 29(2) (2025): 04025009. IF: 2.2.
17. **Bhadane, P.,** Mishra, A. Preparation and Characterization of Graphene Oxide and Pineapple Leaf Fibers Reinforced Rice Starch Bio-Composite. Materials Today: Proceedings (2022). IF: 1.9.

SELECTED CONFERENCE PRESENTATIONS

- **Bhadane, P., Mishra, A.** Effect of Alkaline Treatment of Pineapple Leaf Fibers on the PALF Reinforced Rice Starch Composite. International Conference on Natural Fibers, 2021, Portugal.
- **Bhadane, P., Mishra, A.** Characterization of Pineapple Leaf Fibers and Graphene Oxide Reinforced Rice Starch Bio Composite. International Conference on Novel Materials and Technologies for Energy and Environmental Applications, 2022, India.
- **Bhadane, P., Mishra, A.** Multifunctional Flexible Starch-Graphene Oxide Bio-Composite. MRSI AGM 2023, IIT BHU, and EMRS Spring 2024, Strasbourg, France.

SKILLS

- **Materials & Synthesis:** Metal-organic frameworks, polymer composites, and nanomaterial synthesis; adsorption and separation studies; gas conversion experiments; rare earth element and critical raw material recovery.
- **Characterization:** SEM, FTIR, UV-Vis, DMA, UTM, ICP-OES, TGA-DSC, contact angle goniometry, BET surface area analysis, XRD, XPS, XAS/EXAFS/XANES, NMR, GC-MS/HRMS.
- **Catalysis & Recovery:** Heterogeneous catalysis for CO_2 and CH_4 conversion; heavy metal and REE adsorption kinetics/isotherms; circular economy frameworks for spent-adsorbent valorization.

AWARDS AND FUNDING

- IITGN Overseas Research Experience Fellowship – Visiting Research Scholar, University of Birmingham, UK (6,800 USD).
- Commendation for Outstanding Innovation (PhD), IIT Gandhinagar, 15th Convocation.
- Best Poster Award, University Research Showcase – MOF bead-based water purification system.
- Journal Cover Page Recognition – Environmental Science: Nano and Green Chemistry.
- Gold Medalist – First position, second year of engineering, 9.83/10 CGPA.
- Qualified GATE 2021 with 98.12 percentile.

PATENT

- Bhadane, P., Goyal, P., Mahato, P., Misra, S.K. Adsorbent and a process for recovery of rare earth elements. Indian Patent Office, Application No. 202421047907, filed 21 Jun. 2024.

SELECTED SERVICE

- Assisted in discipline outreach activities for the Department of Materials Engineering, IIT Gandhinagar.
- Certificate of completion in Professional Product Design and Analysis, CADD Centre Training Services, Mumbai.

REFERENCES

- Prof. Abhijit Mishra, Indian Institute of Technology, Gandhinagar — amishra@iitgn.ac.in
- Prof. Superb Misra, Indian Institute of Technology, Gandhinagar — smisra@iitgn.ac.in
- Prof. Kaling Taki, Indian Institute of Technology, Guwahati — kaling.taki@iitg.ac.in
- Dr. Swaroop Chakraborty, University of Birmingham — s.chakraborty@bham.ac.uk